

NATASHA GONSALVES

Hyderabad, India

📞 9529237601 ✉️ [E-mail](#) [LinkedIn](#) [Github](#)

Education

Manipal Institute of Technology

B.Tech. Electronics and Communication Engineering

CGPA - 8.81

2022 – 2026

Manipal, KA

Relevant Coursework

- Computer Architecture
- Electronic Devices
- Digital System Design
- Microprocessors
- Analog Circuits
- Comm. Networks
- MOS Device Modelling
- VLSI Design Flow: RTL to GDS (NPTEL Top 5%)

Experience

IIT Roorkee

May 2025 – June 2025

Research Intern

Roorkee, UK

- Analytical modeling of gate resistance in Nanosheet FETs, focusing on geometry-dependent effects for advanced technology nodes.
- Simulated device characteristics using Sentaurus TCAD and proposed a gate resistance model that captures the bias-dependency of the Nanosheet device.

Manipal Institute of Technology

Sept 2024 – Present

Undergraduate Research Trainee

Manipal, KA

- Exploring Layout development of hybrid magnetic tunnel junction circuits through lambda-based design rules.
- Use of Open-Access tools for IC design from RTL to GDS for a potential tapeout using the sky130 PDK.

Project MANAS

May 2023 – July 2025

Sensing and Automation Subsystem - Member

Manipal, KA

- Implemented UART, I2C, and SPI communication protocols on an STM32 for a sensor suite and safety system, and implemented a watchdog mechanism to trigger on loss of communication
- Designed a two-layer PCB using KiCad for a wireless emergency stop application, housing a buck converter for reduced power loss and compact design.
- Built a motor controller from scratch on an STM32 microcontroller, utilizing encoder values for real-time control.
- Developed a Python Flask application to access a computer's webcam and livestream it on a website with WebSockets for real-time remote bot control.

Projects and Accomplishments

Google Girl Hackathon - Silicon Track

March 2025

- Developed and tested a Machine Learning model for predicting logic depth in SoC design, improving accuracy compared to traditional methods of synthesis report generation.
- Secured 2nd place in Silicon Track among 58000 participants.

Intelligent Ground Vehicle Competition 2024, Michigan, USA

June 2024

- Represented our institution at IGVC '24 in Michigan USA with our UGV *Nova* and bagged the 1st place for Design, 6th in Auto-Nav Challenge and the 4th overall.

Phaser Pedal Design

April 2024

- Developed a phaser pedal using all-pass filters for dynamic sound modulation, incorporating theoretical principles and practical implementation.
- Evaluated performance through subjective tests and DAW processing.

Technical Skills

Languages: C, C++ Verilog

Tools: KiCad, LTSpice, STM32CubeIDE, Sentaurus TCAD, Cadence Virtuoso

Leadership / Extracurricular

IE- Electronics and Communication

Sep 2023 – Present

Technical Head

MIT, Manipal

- Handled the recruitment and on-boarding of junior team members, invited Industry Experts for Tech Talks to help foster interest and curiosity among fellow undergraduates.
- Managed a team of 70+ junior club members for club operations and event, ensuring seamless execution.